

Using Streetlight InSight Data for a Small Area Study: Rickenbacker Area



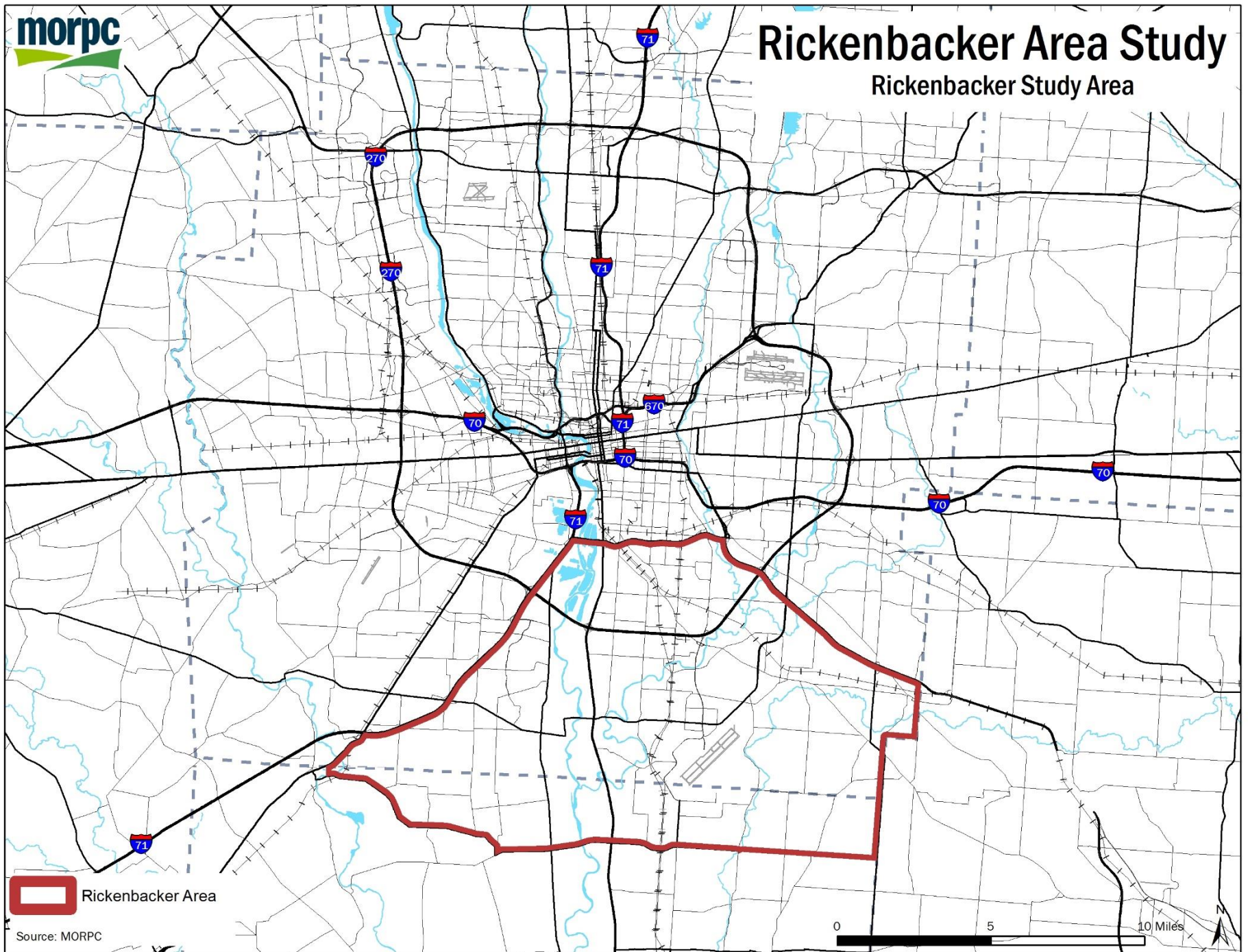
Mid-Ohio Regional
Planning Commission

Background

- Comprehensive study of the area surrounding the Rickenbacker Airport
- Many community stakeholders
- Study in conjunction with the Rickenbacker Airport Master Plan

Rickenbacker Area Study

Rickenbacker Study Area



 Rickenbacker Area

Source: MORPC

0 5 10 Miles

Using Streetlight

- Investigate overall traffic patterns and interaction with the larger region
- Examine seasonal concerns
- Identify freight patterns and access to major routes

Initial Streetlight Project

- Project Type: O-D Analysis (GPS Data)
- Zone Sets:
 - Rickenbacker Study Area Zones
 - External Gates (Pass-Through)
- Options: Default Settings
 - One Year of Data
 - Standard Day Types and Parts

Quick Traffic Summary

Commercial Traffic

	I	E
I	11.58%	26.08%
E	27.11%	35.23%

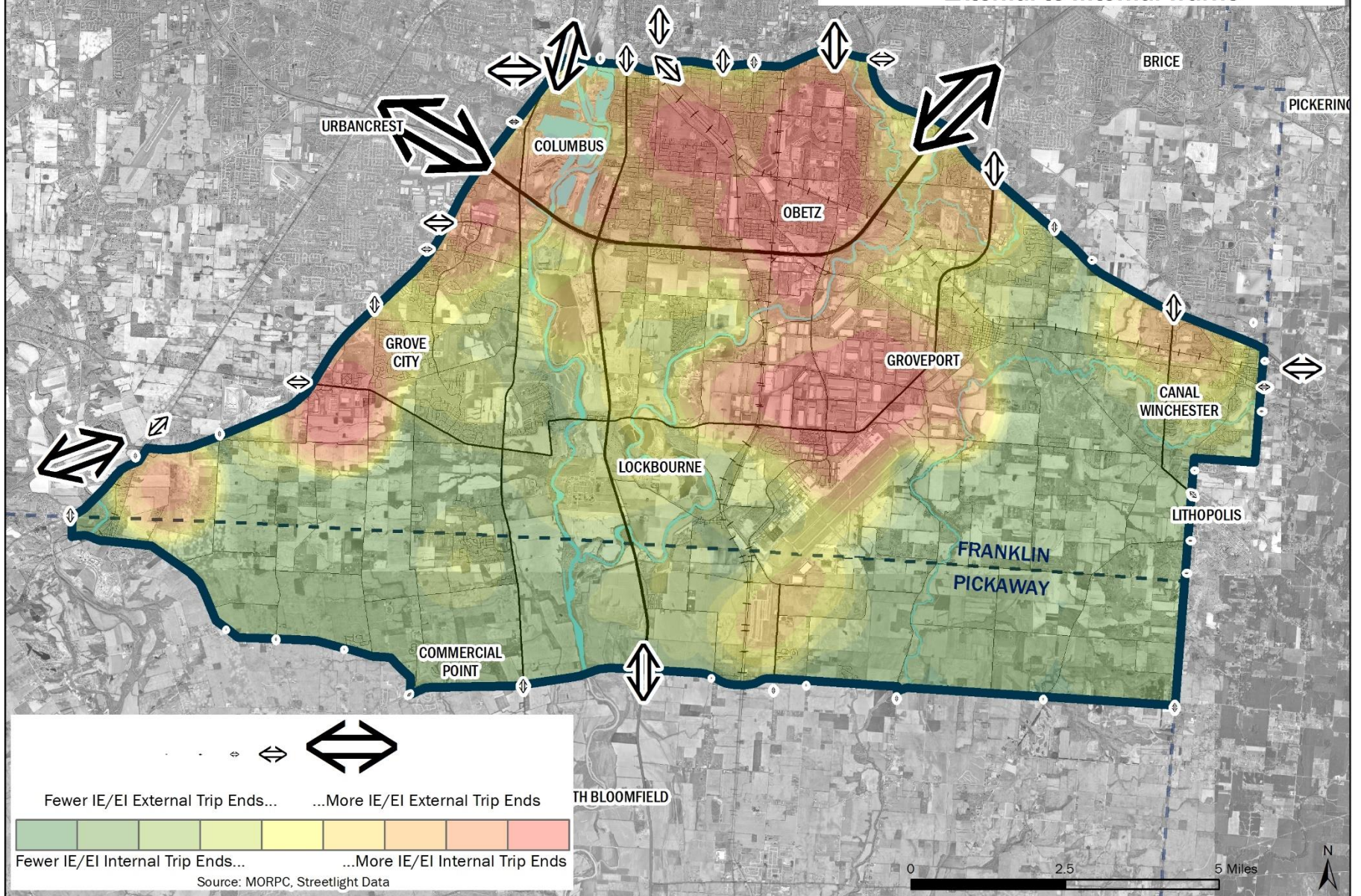
Personal Traffic

	I	E
I	23.25%	27.07%
E	28.58%	21.10%

- Unless otherwise noted all data presented for Average Day, All Day
- Tips & Tricks:
 - Filter out “intra-zonal” traffic for pass through zones
 - Careful with traffic on the “edges”
 - Ultimately ran an O-D analysis with external gates and middle filter to determine true through traffic
 - Due to location inaccuracy buffer used for middle filter

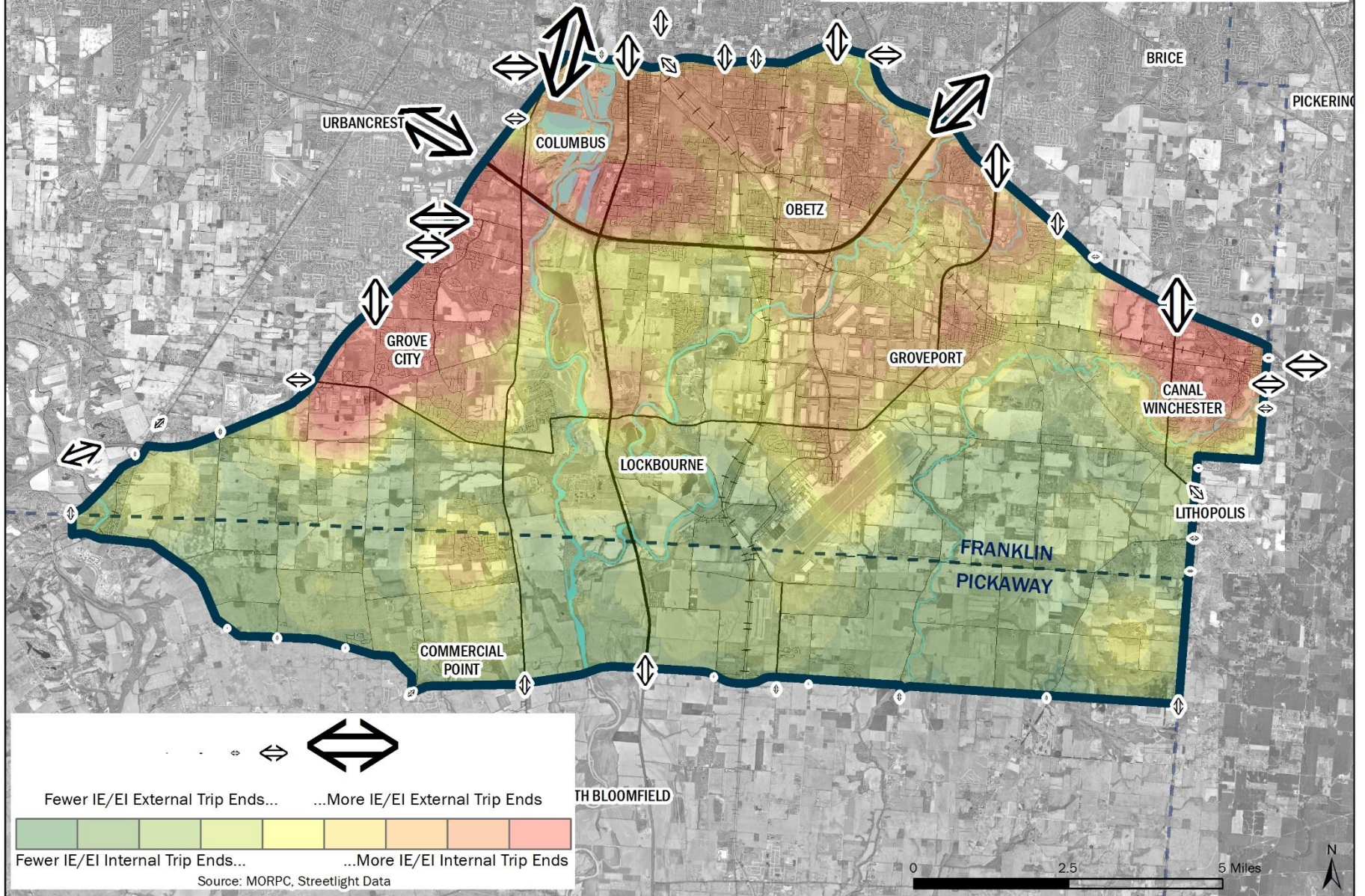
Rickenbacker Area Study

Commercial Internal to External and External to Internal Traffic



Rickenbacker Area Study

Personal Internal to External and
External to Internal Traffic



Seasonal Streetlight Projects

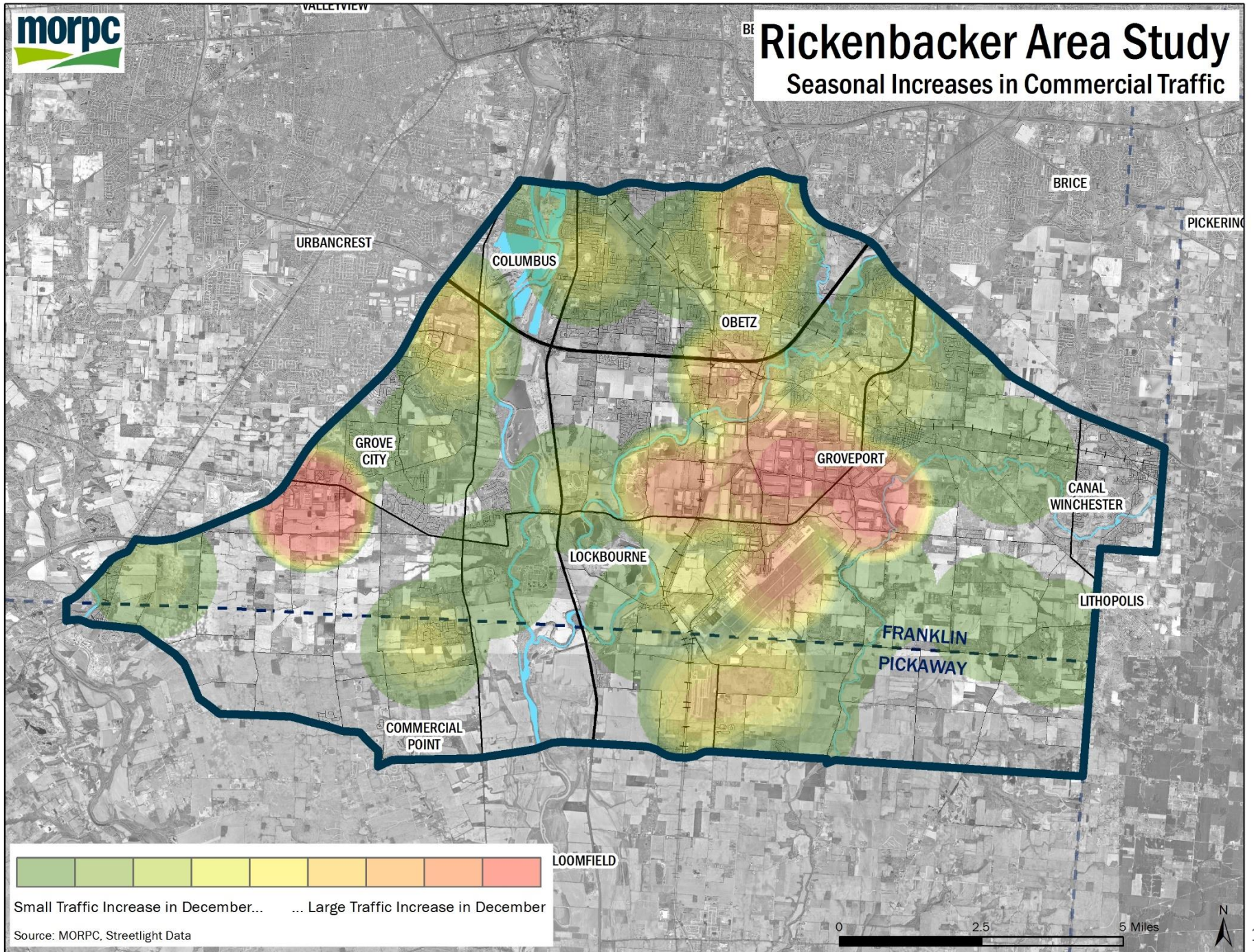
- Project Type: O-D Analysis (GPS Data)
- Zone Sets:
 - Rickenbacker Study Area Zones
 - External Gates (Pass-Through)
- Options:
 - December Run: December for 3 available years
 - Excluding December Run: 11 other months for 3 available years

Seasonal Variations Overall

- Personal O-D flows down 23% in December
- Commercial O-D flows up 1% in December

Rickenbacker Area Study

Seasonal Increases in Commercial Traffic

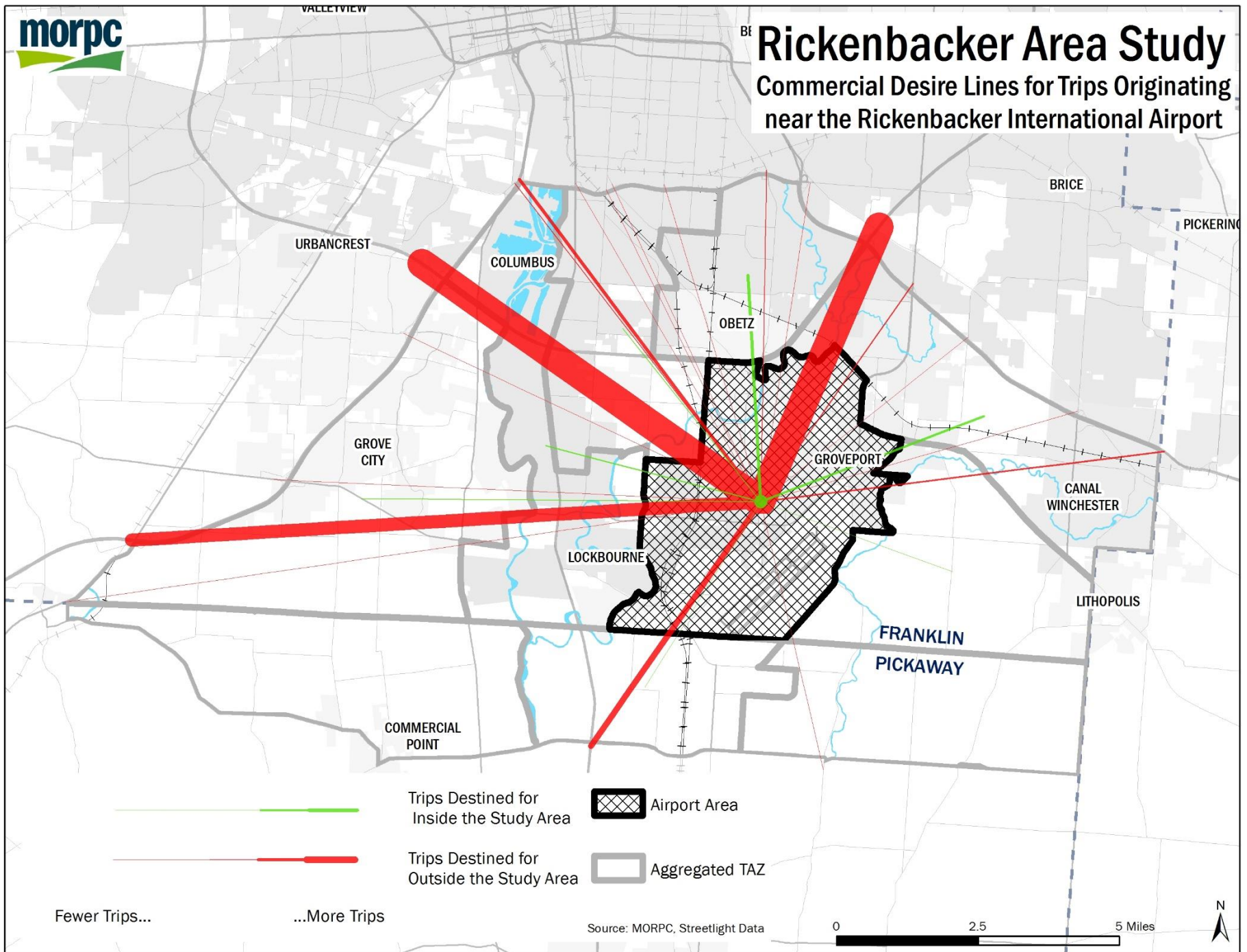


Examining Freight Patterns

- Used initial (full year) data set
- Aggregated Rickenbacker Study Area Zones to a larger geography for presentation purposes

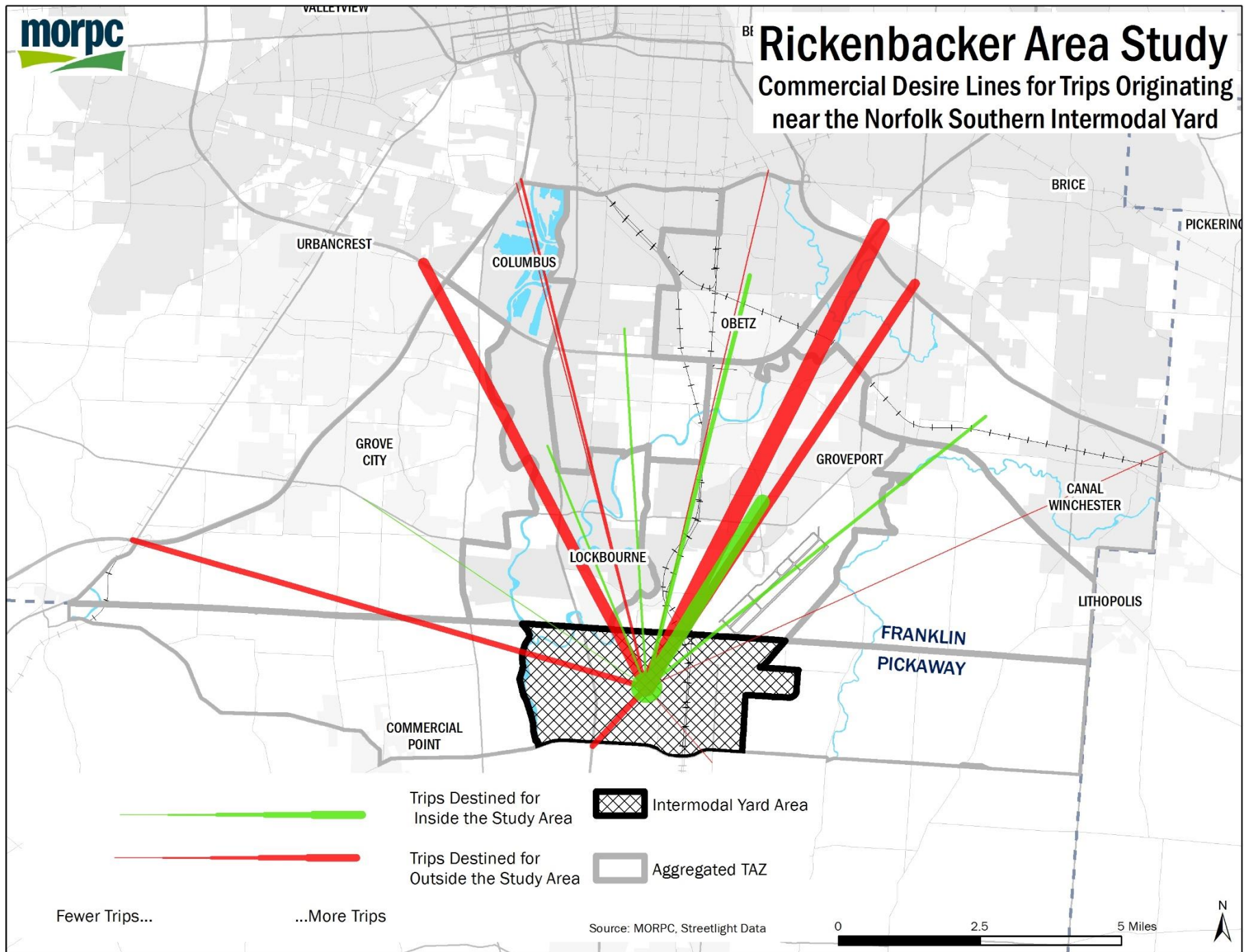
Rickenbacker Area Study

Commercial Desire Lines for Trips Originating near the Rickenbacker International Airport



Rickenbacker Area Study

Commercial Desire Lines for Trips Originating near the Norfolk Southern Intermodal Yard



Examining Access to 71

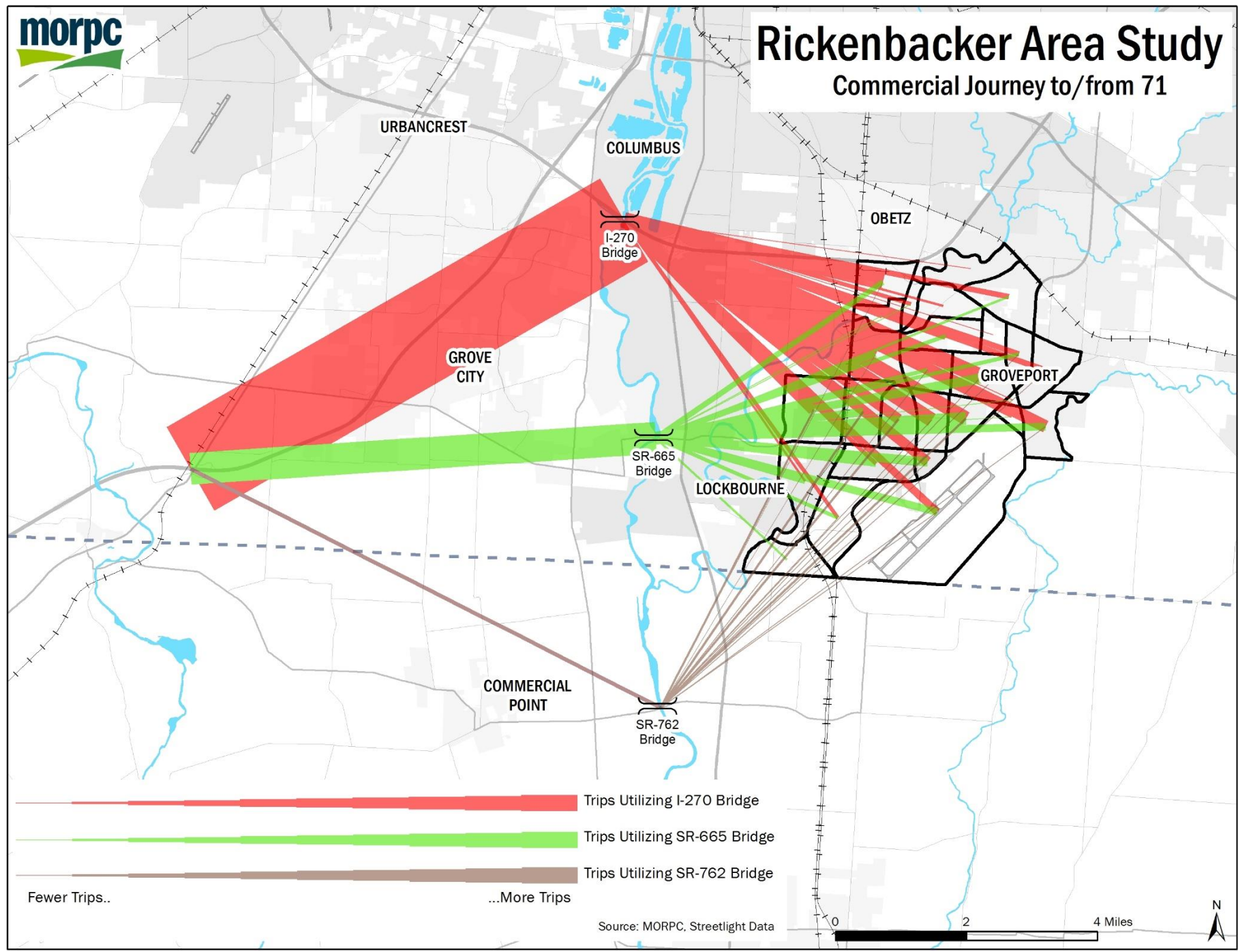
- Some stakeholders concerned with routes from the Rickenbacker Airport area to 71 Southwest of the study area
- Concerns about 665 alignment

Journey to/from 71 Streetlight Project

- Project Type: O-D Analysis with Middle Filter (GPS Data)
- Zone Sets:
 - Rickenbacker Airport Area Zones
 - External Gate to 71
 - 3 Bridge Gates (I-270, SR 665, SR 762)
- Options: Default Settings
 - One Year of Data
 - Standard Day Types and Parts

Rickenbacker Area Study

Commercial Journey to/from 71



Conclusions

- Dataset can be very versatile
- Future Work:
 - Time Period Variation
 - Comparisons/Combination with other Datasets
 - Model Trip Tables
 - LEHD Journey to Work



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